

Master L^AT_EX Exam Cheat Sheet

Prepared for a 50-minute typesetting test

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Contents

1 Fast Preamble Templates

The rule of thumb: load `amsmath` for equations, `graphicx` for images, `xcolor` for colour, `hyperref` LAST. Add `multirow`, `subcaption`, `listings` only when the question needs them.

2 Document Structure

2.1 Title block

Use either `\author` with `\thanks` or the `authblk` package for numbered affiliations:

```
\title{My Title}
\author[1]{Alice}\author[2]{Bob}
\affil[1]{University A}
\affil[2]{University B}
\date{} % remove date
\maketitle
```

2.2 Sectioning & Table of Contents

`\section` `\subsection` `\subsubsection` `\paragraph`. Star versions (`\section*`) are unnumbered and skip the TOC. `\tableofcontents` prints a clickable TOC when `hyperref` is loaded.

3 Text Formatting & Colour

bold, *italic*, underline, `monospace`, `SMALL CAPS`, *emphasis*, `large` and `tiny` sizes. Colour: `red` `text`, `blue text`, `highlighted`, `boxed`. Define your own: `\definecolor{myblue}{RGB}{0,80,180}`.

Size ladder (small $\rightarrow$ big): `\tiny` `\scriptsize` `\footnotesize` `\small` `\normalsize` `\large` `\Large` `\LARGE`

4 Lists & Nested Lists

(a) Ordered with custom label.

(b) Second.

★ bullet with star.

⊕ one-off different marker via `\item[...]`.

I. Roman numerals.

Term definition-style list (bold term + text).

Handy `enumitem` keys: `label=`, `leftmargin=`, `itemsep=2pt`, `noitemsep`, `resume` (continue numbering), `start=3` (begin at 3).

5 Mathematics

5.1 Modes

Inline math with `$...$`: $E = mc^2$. Display with `\[ ... \]` (never use `$$...$$` in modern LaTeX):

$$\int_0^\infty e^{-x} dx = 1.$$

Numbered single equation with `equation`; multi-line with `align` (& = alignment point, `\\` = new line).

5.2 Alignment environments

$$f(x) = (x + 1)^2 \tag{1}$$

$$= x^2 + 2x + 1. \tag{2}$$

`align*` suppresses all numbers; `\nonumber` suppresses one line; `\tag{$\ast$}` forces a custom tag. `gather` centres each line with its own number; `multline` splits one long equation.

$$a = b + c$$

$$d = e - f$$

5.3 Fractions, roots, operators

$$\frac{a}{b} \text{ (display size) vs } \frac{a}{b} \text{ (text size)}, \sqrt{x}, \sqrt[n]{x}, \sum_{i=1}^n i, \prod_k, \lim_{x \rightarrow \infty}, \int_a^b, \oint, \iint, \binom{n}{k}.$$

5.4 Sub/superscripts, deep nesting

Group with braces: x^{2n+1} , $a_{i,j}$. Nest freely: $x_{i_{j_k}}$, $e^{-t/(RC)}$, A^μ_ν (aligned tensor indices via a phantom space), T_M^2 . Multi-line subscript: $\sum_{\substack{i=1 \\ i \neq j}}^n$.

5.5 Delimiters that grow

`\left( \right)` auto-size: $\left(\frac{a}{b}\right)$, $[\]$, $\{ \}$, $| |$, $\langle \rangle$, $\lfloor x \rfloor$, $\lceil x \rceil$. Manual sizes: `\big \Big \bigg \Bigg`.

5.6 Matrices & cases

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad f(x) = \begin{cases} x^2 & x \geq 0 \\ -x & x < 0 \end{cases}$$

`pmatrix=( )`, `bmatrix=[ ]`, `Bmatrix={ }`, `vmatrix=| |`, `Vmatrix=|| ||`, `matrix=none`.

5.7 Spacing in math

Thin $\backslash$, medium $\backslash$: thick $\backslash$; negative $\backslash!$ `\quad \quad \quad`. Text inside math: `\text{...}`. Upright differential: `\mathrm{d}`.

5.8 Symbol quick reference

Greek: $\alpha\beta\gamma\delta\epsilon\zeta\eta\theta\iota\kappa\lambda\mu\nu\xi\pi\rho\sigma\tau\upsilon\phi\chi\psi\omega$ and caps $\Gamma\Delta\Theta\Lambda\Xi\Pi\Sigma\Upsilon\Phi\Psi\Omega$.

Relations: $\leq \geq \neq \approx \equiv \sim \simeq \propto \ll \gg \subset \subseteq \supset \supseteq \notin \ni \prec \succ \preceq \succeq$.

Arrows: $\rightarrow \leftarrow \Rightarrow \Leftrightarrow \Longleftrightarrow \mapsto \longrightarrow \xrightarrow{f} \Uparrow \Downarrow \rightleftharpoons$.

Operators/sets: $\cup \cap \uplus \setminus \times \div \pm \mp \cdot \star \oplus \otimes \odot \wedge \vee \forall \exists \nexists \emptyset \nabla \partial \infty \angle \triangleright \triangleleft \diamond \dagger$.

Blackboard/script/frak: $\mathbb{R} \mathcal{L} \mathfrak{g}$ (need `amssymb/amsfonts`). Dots: $\cdots \vdots \ddots$. Decorations: $\hat{a} \bar{a} \vec{a} \grave{a} \overrightarrow{abc} \overleftarrow{abc} \underline{abc} \overline{abc}$.

6 Tables

6.1 Basic tabular

Left	Centre	Right
a	b	c

`&` separates columns, `\\` ends a row, `\hline` draws a full rule, `\cline{2-3}` a partial rule over columns 2–3.

6.2 multirow + multicolumn + colour

Spanning header		C
Merged	x^2	$s > 0$
	$\frac{1}{s}$	$s > 0$

`\multirow{N}{*}{txt}` spans N rows (* = natural width); `\multicolumn{N}{c|txt|}` spans N columns and re-declares the border; `\rowcolor`, `\cellcolor`, `\columncolor` shade. Math goes in cells simply by using `$...$` or `\dfrac`.

6.3 Line thickness & widths

Thicker rules: `\setlength{\arrayrulewidth}{1pt}` or use `booktabs` (`\toprule` `\midrule` `\bottomrule`). Auto-fit to page: `tabularx` with an X column:

Key	Long wrapping description column
-----	----------------------------------

7 Figures & Graphics

7.1 Single figure

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.6\textwidth]{example-image-a}
  \caption{A caption.}\label{fig:one}
\end{figure}
```

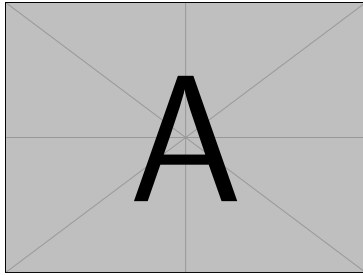
Scaling keys: `width=`, `height=`, `scale=`, `angle=45` (rotate), `keepaspectratio`. Float placement `[htbp]` = here/top/bottom/page; add `!` to override LaTeX's reluctance, or `[H]` (needs `float`) to force exact spot.

7.2 Side-by-side & subfigures

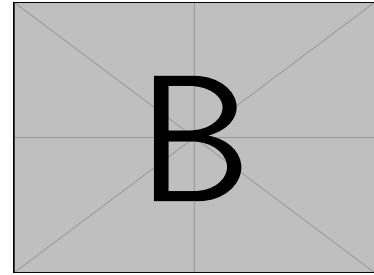
`\ref{fig:sa}` → “??”; `\subref{fig:sa}` gives just the sub-letter. `minipage` works too when you do not want sub-captions: `\begin{minipage}{0.48\textwidth}...\end{minipage}`.

7.3 Wrapped & rotated figures

```
\begin{wrapfigure}{r}{0.4\textwidth} % r=right, l=left
  \centering
  \includegraphics[width=\linewidth]{example-image}
  \caption{Wrapped.}\label{fig:wrap}
\end{wrapfigure}
```



(a) First



(b) Second

Figure 1: Two subfigures.

```
% rotated:
\includegraphics[angle=90]{example-image} % rotate the image
\begin{sidewaysfigure}...\end{sidewaysfigure} % rotate whole float (rotating)
```

8 Cross-References & Hyperlinks

Label anything with `\label{type:name}` directly after its `\caption` or in the equation, then cite with `\ref` (number), `\eqref` (adds parentheses for equations), `\pageref` (page), or `\autoref` (adds the word “Figure”/“Table”). External link: `\href{https://url}{clickable text}`; bare URL: `\url{https://url}`. Configure colours with `\hypersetup{...}`. A clickable TOC comes free once `hyperref` is loaded.

9 Code Listings

Listing 1: Demo

```
1 def greet(name):
2     """Say hello."""
3     return f"Hello, {name}" # f-string
```

Inline code: `\lstinlineprint(x)|`. From a file: `\lstinputlisting[style=code]{script.py}`. For prettier output use `minted` (needs `-shell-escape` + `Pygments`).

10 Theorems, Proofs & Custom Boxes

Definition 10.1. A statement being defined.

Theorem 10.1 (Named). *The claim.*

Proof. Argument. □

Proposition 10.2. *Shares the theorem counter (see the preamble `[theorem]` option).*

A coloured “custom theorem box” with `tcolorbox`:

Key Result

$E = mc^2$ inside a coloured, titled box.

11 Algorithms / Pseudocode

Algorithm 1: Find maximum (algorithm2e)

Input: array A of length n
Output: maximum element

```
1  $m \leftarrow A[1];$ 
2 for  $i \leftarrow 2$  to  $n$  do
3   if  $A[i] > m$  then
4      $m \leftarrow A[i];$ 
5   end
6 end
7 return  $m;$ 
```

Alternative package: `algorithm + algpseudocode` (`\State` `\For` `\While` `\If` `\EndIf`).

12 Footnotes

Text with a footnote.¹ Multiple footnotes auto-number. Reuse one with `\footnotemark` / `\footnotetext`.

13 References / Bibliography

Manual (no external tool, fastest for exams):

```
\begin{thebibliography}{9}
  \bibitem{key1} A. Author, \textit{Title}, Publisher, Year.
  \bibitem{key2} B. Writer, ‘‘Article,’’ Journal, 2024.
\end{thebibliography}
```

Cite in text with `\cite{key1} → [1]`. The `{9}` argument just sizes the label box (one digit here).

14 Predicted Tricky Variations

1. **Aligned multi-column equations:** use `alignat` to control inter-column spacing manually: `\begin{alignat}{2} a&=b \quad c&=d \end{alignat}`.
2. **Piecewise + annotations:** cases plus `\underbrace{expr}_{\text{note}}` and `\overbrace{expr}^{\text{note}}`.
3. **Numbered sub-equations:** wrap several equations in `\begin{subequations} ... \end{subequations}` to get (1a),(1b).
4. **Long division-style derivations:** keep `&=` aligned and end intermediate lines with `\nonumber` so only the final result is numbered.
5. **Units:** load `siunitx` and use `\SI{9.8}{m/s^2}` and `\num{1.496e11}` for clean scientific notation.
6. **Coloured full-row headers:** `\rowcolor` must be the FIRST token of the row, before any `\multicolumn`.
7. **Landscape table/figure:** `{sidewaystable}` or the `lscape {landscape}` environment.

¹This is the footnote body.

8. **Custom theorem box with counter:** `tcolorbox` with `\newtcbtheorem` gives numbered, coloured theorems.
9. **Chemical/tensor indices:** pre-subscripts via `{\_a^b}` and aligned indices via `phantom` or `\vphantom`.
10. **Boxed final answer:** `\boxed{x = \frac{-b \pm \sqrt{\Delta}}{2a}}`.

Referenced items for completeness: Equation (??), Theorem ??, Figure ??, Listing ??.